Trace the evidence. Make the review count.

An investigation workspace that connects deterministic alerts, validated AI explanations and accountable human decisions.

Draft for review - 17 October 2026

Local implementation verified. Genuine AML investigation, human review and deployment evidence are pending.

THE PROBLEM AND INTENDED VALUE

Alerts need a reviewable explanation.

An analyst must connect transactions, rule signals and case context before deciding what deserves further investigation.

Workflow problem

Evidence, explanations and review decisions can become disconnected. An explanation without provenance adds another thing to verify.

Product hypothesis

Keep the evidence and review together. Show what is known, what failed and which human decision supports an evaluation.

Target user: an AML analyst reviewing synthetic cases. Faster review and better precision are hypotheses to measure; no customer, productivity or financial outcome is claimed.

INVESTIGATION TO REVIEW

One sealed step before the next.

A1

Deterministic evidence enrichment

A2

AI explains code-owned risk arithmetic

A3

AI categorizes with bounded evidence

A4

AI drafts the private investigation brief

A0

Deterministic prioritization after validation

A5

Separate analysis of genuine human feedback

Schema, citation and protected-signal checks gate completion. A failure stops downstream execution and stays visible to the operator.

No genuine completed AML investigation is stored yet. Test failures and display fixtures do not count as successful AI output or human feedback.

ARCHITECTURE AND BOUNDARIES

Public evidence. Private investigation.

Public web and API

Frozen deterministic cases and aggregate usage. No model narrative, private review or held-out labels in public reads.

React + FastAPI / signed-out journey

Operator and admin API

Private reads, explicit enqueue and refresh, review history, immutable configuration and attempt ledger.

Firestore / separate operator and admin allowlists

Separate private worker

Named Cloud Tasks delivery, authenticated invocation, 300-second lease and 240-second execution bound.

Prepared code / real cloud delivery unverified

Shared model gateway

Explicit Developer or cloud transport. Exact input counting, atomic reservations, validators and measured usage.

Gemini SDK / cloud execution disabled

CONTROLS THAT MAKE FAILURE VISIBLE

Bound requests. Preserve uncertainty.

Before a request

- Recheck mode, credits, expiry, switch, cap and worker lease.
- Reserve each attempt atomically.
- Freeze model, prompts and retry limits in immutable versions.

After a request

- Seal the attempt before the next leg.
- Permit one retry only for a known short transient rejection.
- Keep unknown provider outcomes, usage and cost visibly unknown.

Input / generated-token caps: A2 4096 / 1536; A3 2048 / 768; A4 4096 / 2048; A5 8192 / 3072. Cloud daily maximum: 120 attempts, including retries. New UTC days start disabled.

A linked billing account does not establish credit coverage. No cloud deployment or additional live model call was made in this preparation batch.

MEASURED LOCAL VERIFICATION

What the current evidence establishes.

43

Unit tests passed

9 emulator-only checks run separately.

9

Integration tests passed

Fresh local Auth / Firestore emulators. Zero integration skips.

112

Frozen synthetic cases

40 discovery, 20 selection, 40 held-out, 12 public demo. Nine detector rules.

Checks cover immutable publication, duplicate enqueue, access boundaries, real-ledger fault accounting, fenced delivery and refusal without fabricated model output.

One earlier authorized Developer API connectivity smoke reported 74 tokens. It was not an AML investigation; cost remains unknown. Recorded dependency audits: zero production npm and installed Python findings; eight development-tool advisory entries remain for release review.

`Provenance: docs/evaluation-record.json - local worktree fingerprint and actual JUnit results; release_ready = false.`

EVOLUTION WITHOUT INVENTED EVIDENCE

Human reviews define the denominator.

Sufficiency is computed

A supported detector proposal requires 30 conclusive reviews overall and 15 for the target rule. A5 can return INSUFFICIENT_EVIDENCE honestly.

Freeze before held-out access

The operator explicitly freezes an allowlisted candidate. Its hash precedes one stored, idempotent held-out simulation. Nothing activates automatically.

Measure analyst review time, conclusive-review counts, removed benign and suspicious alerts, and alerts retained by other rules.

Current genuine feedback, stored A5 and held-out comparison evidence are absent. No precision gain, time saving or detector improvement is claimed.

RELEASE PATH

Close the evidence gaps, then publish.

- Verify funding, model / location, runtime identities and service coverage.
- Complete one genuine investigation and manually ground its brief.
- Capture your analyst verdict, real A5 and explicit candidate freeze; store the held-out comparison.
- Build and deploy approved artifacts; verify signed-out and operator journeys.
- Finalize the public repository, PDF and a video under three minutes.

This file is a local draft. No public demo, repository, video or submission URL is claimed. Replace draft evidence gaps only when actual stored results and deployed revision IDs exist.